AncholVox: A Silver-Standard Diagnostic Benchmark for Multi-Dialect Bengali and Bengali–English Code-Switched Speech Recognition

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Abstract—We introduce AncholVox, a silver-standard diagnostic benchmark for multi-dialect Bengali ASR and Bengali–English code-switching detected in automatic transcripts. It contains 41,499 clips (131.2 hours) from 15 regional labels, constructed through speech segmentation, Whisper large-v3 transcription, script-based switch detection, and text-only LLM filtering. On the archived historical split, fine-tuned wav2vec2 achieves the lowest primary WER against the silver references (0.861). These model comparisons are diagnostic because that split is not source-disjoint. In a limited 100-clip, two-annotator validation study, MMS-1B has the lowest WER against both human references (0.757 and 0.697), showing that rankings depend on the reference. AncholVox is therefore presented as an auditable dataset and stress-test benchmark for silver-label noise, dialect variation, decoder repetition, and English-token retention, not as a new state-of-the-art ASR model.[1]

Index Terms—Automatic speech recognition, Bengali ASR, code-switching, dialectal speech, low-resource speech recognition, Whisper, and a large language model judge.

I. Introduction

Automatic speech recognition (ASR) does not work equally well across languages, regional varieties, and speaking styles. Dialect variation and code-switching (CS), in which speakers alternate between languages within an utterance, are particularly challenging when annotated speech is scarce.

Bengali has substantial regional variation, including Chittagonian, Sylheti, Noakhali, Rangpuri, and Barishali varieties, which differ in pronunciation, vocabulary, and expression. Informal online speech may also contain Bengali and English words in the same utterance. For example, a speaker may say, 'chatgaiya bhashay phone unboxing kortachi, honestly speaking camera performance onek bhalo," meaning 'unboxing a phone in the Chittagong dialect; honestly speaking, the camera performance is very good."

Despite this, most existing Bengali speech datasets focus on only one problem at a time. For example, read-speech datasets like OpenSLR-37 [1] mainly cover standard Bengali. The Ben-10 corpus [2] includes ten regional dialects but does not include code-switching. On the other hand, the MUCS shared task [3] provides Bengali–English code-switching data, but only in standard Dhaka Bengali. We are not aware of another released resource at this scale that jointly targets broad regional variation and Bengali–English switching detected in silver transcripts. This gap between real-world speech and available resources makes it difficult to study robust ASR systems.

Recognizing dialectal code-switched speech is difficult for several reasons. First, regional dialects can be quite different from standard Bengali in pronunciation, sound patterns, rhythm, and vocabulary, so models trained on standard Bengali do not perform well. Second, code-switching requires the system to handle smooth switching between Bengali and English within the same sentence, often mixing languages that standard monolingual models are not designed for. Third, there is very little training data for dialectal and code-switched speech, and manually transcribing data across 15 dialects is very expensive. Finally, many dialect words and borrowed English terms do not have fixed spellings, so even correct transcriptions may be marked wrong by strict evaluation metrics. These challenges together motivate the design of our dataset, pipeline, and evaluation approach. They lead us to build an automated annotation pipeline without manual transcription, use script-aware code-switch detection, adopt a shared vocabulary sequence-to-sequence model, and focus on character-level and code-switch-aware evaluation metrics.

This work fills this gap; Fig. 1 provides an end-to-end overview of the dataset construction, models, and evaluation. The main contributions of this paper are as follows:

1) One of the first large-scale resources designed specifically for the intersection of regional Bengali dialects and Bengali–English switching detected in silver transcripts: 41,499 clips (131.2 hours) across 15 regional labels.
2) A scalable automated construction pipeline using voice-activity segmentation, Whisper-based silver transcription, script-based code-switch detection, and text-only multilingual LLM review of borderline cases, complemented by a two-annotator validation subset.
3) A diagnostic benchmark comparing three ASR model families under zero-shot and fine-tuned settings, including Bengali-only, code-switch-only, and joint-data

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[1]Dataset: https://huggingface.co/datasets/niloycste68/Bangali_local_dialect_ASR_HF_Dataset — gated access pending the institutional determinations recorded in the release register; code: https://github.com/niloycste/ASR-Bengali-Local-Dialect-.

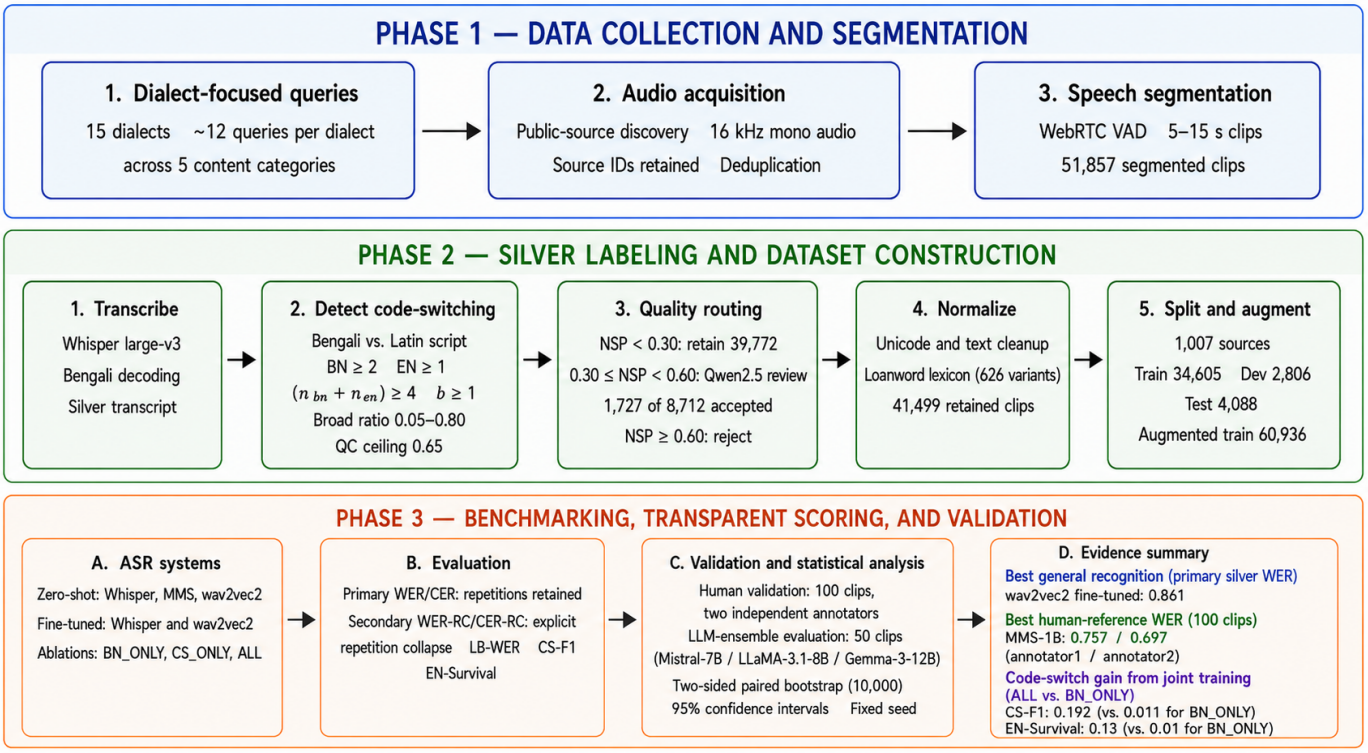


Fig. 1. Auditable overview of dataset construction and evaluation. Clips with no-speech probability below 0.30 are retained, clips in $[0.30, 0.60)$ receive LLM review, and clips at or above 0.60 are rejected. Evaluation separates conventional surface-normalized scores, which retain repetitions, from explicitly labeled repetition-collapsed system scores.

training recipes. Because these recipes differ in coverage and data budget, they provide practical comparisons rather than a controlled causal estimate of each data type.

4) A detailed evaluation and error analysis across dialects and code-switching conditions, including two-sided significance tests, bootstrap uncertainty for code-switching metrics, and an LLM-ensemble meaning-preservation diagnostic.

AncholVox is positioned primarily as a dataset-construction and auditability contribution. The model experiments are diagnostic stress tests of the released resource, not an exhaustive state-of-the-art leaderboard. Their purpose is to measure failure modes under a common protocol and to separate model behavior from silver-reference noise, which is characterized on a limited sample in Section III-E.

The rest of the paper is organized as follows: Section II reviews related work on Bengali, dialectal, and code-switched ASR. Section III describes the AncholVox dataset and its construction process. Section IV presents the ASR models and fine-tuning strategy, while Section V details the experimental setup. Section VI reports the main results, Section VII provides detailed error analysis, training-recipe comparisons, and case studies, Section VIII discusses limitations and reproducibility, and Section IX concludes the paper with future directions.

II. Related Work

This section reviews prior studies on Bengali automatic speech recognition (ASR), covering standard and dialectal ASR, code-switched speech recognition, large scale pretrained speech models, automated speech corpus construction, and large language model (LLM)-based data quality assessment. While recent advances in multilingual and low-resource ASR have improved speech recognition performance across many languages, existing Bengali ASR research has largely focused on standard Bengali speech. As a result, the combined challenges of regional dialect variation and Bengali–English code-switching remain underexplored.

A. Bengali and Dialectal ASR

Early Bengali ASR systems were mainly based on Gaussian mixture models and deep neural network hidden Markov models with manually designed pronunciation lexicons [4], [5]. These systems mostly focused on standard Bengali and had limited ability to handle unseen words. Later, Bengali ASR research moved toward end to end approaches, including connectionist temporal classification (CTC) [6], deep recurrent networks [7], Deep Speech [8], and attention based encoder decoder models such as Listen, Attend and Spell [9]. Toolkits such as ESPnet [10] further supported this progress and helped reduce error rates for standard Bengali ASR [11]. More recently, the Trans-

former [12] and masked pre training [13] have enabled powerful self supervised and weakly supervised speech models, including wav2vec,2.0 [14], HuBERT [15], WavLM [16], XLS-R [17], [18], Whisper [19], and MMS [20]. Together with multilingual datasets such as Common Voice [21], these models have improved transfer learning for resource limited Bengali speech recognition. However, dialectal Bengali has received focused attention only recently. The Ben-10 corpus [2], which contains more than 78 hours of speech from 394 speakers across ten regional dialects, shows that even strong foundation models such as Whisper large v3 perform poorly on dialectal speech that differs from the training data, both before and after fine tuning. Biswas et al. [22] propose a WavLM based denoising and adaptation framework for dialectal Bengali, including Chittagonian and Sylheti, under simulated noise, and report large relative reductions in WER compared with wav2vec,2.0 and Whisper at low signal to noise ratios. Ridoy et al. [23] compare Whisper and Wav2Vec-BERT on standard Bangla and find that the self supervised encoder is both more accurate and more computationally efficient in resource limited settings. However, these studies do not address code switching, which motivates the need for a benchmark that brings dialect variation and code switching together in a single resource.

B. Code-Switching ASR

Code switching, where speakers alternate between two languages within a conversation or even within a single sentence, is common in multilingual communities. It is especially frequent in informal Bangladeshi speech, where English words are often inserted into Bengali sentences. This creates several challenges for ASR systems. First, the model must recognize words from two languages and, in the case of Bengali and English, two different writing systems. Second, code switched speech is still rare in standard ASR training datasets. Third, language switching often occurs together with accent, pronunciation, and acoustic variation, which makes both speech recognition and language modeling more difficult [24], [25]. Progress in code switched ASR has largely been supported by dedicated datasets, such as the Mandarin English SEAME corpus [26]. However, recent studies show that even large multilingual models do not handle code switching reliably unless they are trained with explicit code switched data [27]. For Indian languages, the MUCS 2021 challenge [3] introduced Bengali English and Hindi English code switching benchmarks, but the Bengali data is limited to standard Dhaka Bengali and does not include regional dialects. One useful feature of Bengali English code switching is that the two languages use different Unicode ranges. This allows word level language identification to be performed almost directly from the script, without requiring a separate language identification model. From a modeling perspective, sequence to sequence models such as Whisper are better suited to this setting because their shared subword vocabulary [28] can represent both scripts and generate mixed script text in a single decoding process. In contrast, monolingual CTC models such as MMS are usually restricted to one output script and therefore cannot properly represent code switched speech. This work evaluates both modeling paradigms to measure this gap in dialectal Bengali ASR.

C. Automated Corpus Construction for Low-Resource ASR

Since manual transcription is difficult to scale for many low-resource language varieties, recent speech corpora increasingly rely on automated data construction pipelines. For example, GigaSpeech 2 [29] builds a large multi-domain corpus for Thai, Indonesian, and Vietnamese using automated web crawling, Whisper-based transcription, and iterative refinement. This shows that weakly supervised labels can support large-scale corpus creation without relying entirely on human transcribers. Other studies have also used synthetic speech and spectrogram augmentation to increase the amount of training data for low-resource ASR [30]. AncholVox follows this automated approach, but focuses on a previously underexplored setting: multi-dialect Bengali speech with English code switching. It also adds an LLM-based judge to filter borderline or uncertain transcripts.

D. LLM-as-a-Judge for Data Quality

Large language models (LLMs) are increasingly used as automatic evaluators in natural language processing, including pairwise response judging [31], [32] and multi-dimensional quality scoring [33]. These models are especially useful when human annotation is expensive and reference labels are limited or noisy, as in low-resource dialectal ASR. However, their reliability depends on careful prompt design and the capability of the judging model. In this work, we use LLM judges at two stages. During dataset construction, a strong multilingual model, Qwen2.5 [34], evaluates borderline silver-standard transcripts across several quality dimensions and decides whether they should be retained. During evaluation, an ensemble of three open models, Mistral [35], LLaMA [36], and Gemma [37], provides an additional reference-free assessment that complements word-level metrics, as discussed in Section VII. To the best of our knowledge, using LLM judges for both quality control and evaluation in code-switched low-resource ASR has not been previously explored.

Qwen2.5 was selected because its open weights permit fixed, local inference: transcripts need not be sent to a commercial API, and the same model and prompt can be rerun by other researchers. This choice is operational rather than a claim that Qwen2.5 is superior to commercial judges or manual review. Moreover, the construction-stage judge receives text only; it assesses transcript plausibility and cannot verify acoustic alignment. The threshold cascade and the independent human sample are therefore

essential safeguards rather than treating the LLM decision as ground truth.

Recent multilingual systems provide additional context for the experimental scope. SeamlessM4T supports Bengali as an input speech language and is a relevant future cross-architecture baseline [38]. Canary-1B is a strong multilingual ASR model, but its released language set is limited to English, German, French, and Spanish and therefore does not provide a Bengali decoding condition [39]. The present suite is consequently representative rather than exhaustive.

E. Comparison with Existing ASR Resources

Table I summarizes representative Bengali and code-switching ASR resources based on the dimensions most relevant to this study. The comparison highlights two clear gaps in the existing literature. First, standard Bengali ASR corpora and models [1], [4], [5], [11] mainly focus on standard speech and do not address either regional dialect variation or code switching. Second, existing specialized resources cover only one of these challenges at a time. The Ben-10 corpus [2] includes multiple Bengali dialects but does not contain code-switched speech, while the MUCS task [3] provides Bengali English code-switching data only for standard Bengali. Although large multilingual models such as XLS-R, Whisper, and MMS [17], [19], [20] support transfer to Bengali, their performance often drops on dialectal speech. In addition, monolingual CTC models cannot properly generate mixed-script code-switched output. AncholVox addresses this gap by focusing on the intersection of these two underexplored problems: multi-dialect Bengali speech and Bengali–English switching detected in silver transcripts. To the best of our knowledge, it is one of the first resources designed specifically for this setting.

III. The AncholVox Dataset Construction

AncholVox is designed to study regional Bengali speech and Bengali–English switching detected in automatically generated transcripts. It combines dialect-targeted social-media collection with automatic segmentation, transcription, script-based labeling, and quality filtering. The search design increases the likelihood of retrieving informal regional speech, but neither a query label nor a silver transcript independently verifies a speaker's dialect or the words present in the audio. AncholVox is therefore presented as a large, auditable diagnostic dataset rather than a manually verified corpus.

A. Dataset Design Principles

AncholVox is built on three principles. First, dialect-name queries are used to increase regional coverage because region-name queries, such as "Chittagong vlog," may retrieve standard Bengali intended for a broad audience. Second, genres likely to contain informal Bengali–English mixing are prioritized, and switching labels are subsequently derived from the silver transcripts. Third, the construction pipeline is automated so that the corpus can be expanded without requiring manual transcription of every clip. These principles guide collection; they do not replace acoustic or speaker-level verification.

B. Dialect-Aware Search Query Design

Search queries determine which creators and speech styles are retrieved. AncholVox therefore combines dialect-name queries with content genres likely to contain informal speech and English terms. For each regional label, around twelve queries are designed across the five categories in Table II. Street interviews and live reactions tend to contain unscripted responses, while technology, gaming, campus, and career content often contains English terminology. Vlogs add relaxed conversational speech. Videos uploaded before 2021 and videos with very low viewership are excluded.

C. Construction Pipeline

Algorithm 1 summarizes the end-to-end pipeline; each step is justified below.

Step 1: Data Acquisition and Segmentation: Audio is collected from social media platforms, including YouTube, TikTok, and Facebook, using yt-dlp. All audio is downloaded in 16 kHz mono WAV format. A shared de-duplication archive is used to ensure that the same video is not collected more than once. Since social media recordings often contain long silent or non-speech parts, each recording is divided into shorter 5–15 s speech clips using WebRTC voice activity detection. This step produces the audited total of 52,020 clips (165.23 hours). Language filtering after transcription retains 48,772 clips (155.08 hours), of which 48,484 fall below the upper no-speech threshold and enter the confidence cascade described below.

Step 2: Transcription: Each clip is transcribed using Whisper large-v3 [19], with the language setting forced to Bengali. It is used as a reproducible, openly available multilingual labeler; its output remains a noisy substitute for manual transcription in this difficult setting.

Step 3: Code-Switch Detection: Code-switching is detected using Unicode script information. Bengali words use the Bengali Unicode range (U+0980–U+09FF), while English words usually appear in Latin script. Since these scripts are separate, word-level language identification can be performed almost directly from the script, without using a separate language identification model.

Let n_{bn} be the number of Bengali-script words, n_{en} be the number of Latin-script words, b be the number of language boundaries between adjacent words of different scripts, and $\rho = n_{\mathrm{en}}/(n_{\mathrm{en}}+n_{\mathrm{bn}})$ be the English-word ratio. A clip is labeled as code-switched if it satisfies the following conditions:

$$\begin{aligned}(n_{\mathrm{bn}} + n_{\mathrm{en}}) \geq 4, \quad n_{\mathrm{bn}} \geq 2, \quad n_{\mathrm{en}} \geq 1,\\ b \geq 1, \quad 0.05 \leq \rho \leq 0.80.\end{aligned} \tag{1}$$

A stricter high-confidence subset is also created by requiring $n_{\mathrm{en}} \geq 2$ and $0.10 \leq \rho \leq 0.70$. During automatic

TABLE I
Summary of representative Bengali and code-switching ASR resources and models. "CS" denotes code-switching support, and "–" indicates not applicable. AncholVox is, to our knowledge, one of the first resources to combine multi-dialect Bengali coverage with English switching detected in silver transcripts.

Ref.	Method / Model	Objective	Variety	CS	Hours	Key limitation
Hasnat et al. [4]	GMM-HMM	Isolated-word Bengali ASR	Standard BN	No	–	Lexicon-dependent; WER >50%; no OOV handling
Swarna et al. [5]	Survey (HMM/DNN)	Bengali phoneme recognition survey	Standard BN	No	–	Survey; standard Bengali, no dialect or CS
Mandal et al. [11]	End-to-end (CTC/attention)	Lexicon-free Bengali ASR	Standard BN	No	–	Standard Bengali only; no dialect or CS
Kjartansson et al. [1]	OpenSLR-37 corpus	Read-speech benchmark	Standard BN	No	100	Read speech; no dialect, no code-switching
MUCS 2021 [3]	CS challenge corpus	Bengali–English CS-ASR	Standard BN	Yes	16	Standard Dhaka Bengali; no dialect
Ben-10 [2]	Whisper fine-tuning	Multi-dialect Bengali ASR	Dialectal BN	No	78	No code-switching
XLS-R [17]	wav2vec2 self-supervised	Multilingual transfer	Multilingual	No	–	Standard Bengali via transfer; no dialect/CS
Whisper [19]	Weakly-supervised seq2seq	Multilingual ASR	Multilingual	No	–	∼70% WER on unseen Bengali dialects
MMS [20]	CTC with adapters	1000+ language ASR	Multilingual	No	–	Monolingual output; cannot emit code-switching
Ridoy et al. [23]	Whisper vs. Wav2Vec-BERT	Low-resource Bangla model comparison	Standard BN	No	90–100	Standard Bangla only; no dialect or CS
Biswas et al. [22]	WavLM denoising + adaptation	Noise-robust dialectal Bengali ASR	Dialectal BN	No	∼20	Dialectal but no code-switching; noise-focused
GigaSpeech 2 [29]	Automated crawl + transcribe + refine	Large-scale low-resource corpus	Th/Id/Vi	No	∼30k	Not Bengali; no dialect or code-switching

TABLE II
Content angles used to design dialect-name search queries and why each is likely to contain informal speech or Bengali–English mixing.

Content type	Reason for dialectal code-switching
Technology/phone review	Frequently contains English technical terms
Street interview/reaction	Often contains unscripted public responses
Gaming / live stream	Live commentary with frequent English game terms
Campus/career	Academic and job English mixed into regional speech
Vlog / content creator	Relaxed informal speech with frequent CS

annotation, an additional conservative QC check excludes CS candidates with $\rho > 0.65$ from automatic acceptance. Thus, 0.80 is the detector's broad labeling boundary, whereas 0.65 is the release-stage automatic-acceptance ceiling. The broad detector rule is applied to model outputs when computing CS-F1.

Step 4: Silver-Standard Annotation with an LLM Judge: Since large-scale manual transcription of dialectal code-switched speech is not practical, AncholVox uses a two-stage silver-standard annotation strategy. We define $\mathrm{NSP}(c)$ as the Whisper large-v3 no-speech probability for clip c, used as a confidence proxy for speech presence and transcript reliability. Clips with $\mathrm{NSP}(c) < \tau_{\mathrm{lo}} = 0.30$ are retained automatically as high-confidence speech, following prior evidence that high-confidence Whisper transcripts can provide useful training labels [29]. Clips with $\mathrm{NSP}(c) \geq \tau_{\mathrm{hi}} = 0.60$ are rejected as low-confidence or non-speech clips. The remaining clips, with $0.30 \leq \mathrm{NSP}(c) < 0.60$, are treated as borderline cases.

For the borderline set, Qwen2.5 [34] evaluates each transcript using the same local Ollama-based prompt. Open weights make the decision rule reproducible while keeping transcripts off third-party APIs. The prompt asks four yes/no questions: whether the transcript is valid Bengali rather than hallucinated text, whether Bengali–English mixing is natural, whether it resembles dialectal or informal spoken Bengali, and whether it resembles speech rather than noise, repeated text, or song lyrics. The implementation returns accept when three or four criteria pass, uncertain for a two–two split, and reject when zero or one criterion passes. The parser derives this outcome from the four criterion values rather than trusting the model's self-reported overall label. Only accept clips enter the release.

The stored pre-judge and final manifests make the cascade auditable. Retaining only NSP < 0.30 would yield 39,772 clips, whereas accepting every borderline candidate would yield 48,484. Qwen2.5 retains 1,727 of the 8,712 borderline clips (19.8%) and excludes 6,985 (80.2%), giving the 41,499-clip final set. These counts quantify the

Algorithm 1 AncholVox: From raw dialectal audio to a curated CS-ASR benchmark

Input: dialect-name queries $\mathcal{Q}$; VAD aggressiveness a; no-speech thresholds τ_{lo}=0.30, τ_{hi}=0.60; historical split target $\approx$ 83/7/10
Output: curated dataset $\mathcal{D} = (\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{dev}}, \mathcal{D}_{\text{test}})$
Step 1: Acquire and segment
$\mathcal{R} \leftarrow$ Download16k($\mathcal{Q}$) // yt-dlp + ffmpeg; shared de-duplication archive
$\mathcal{C} \leftarrow$ VADSegment($\mathcal{R}, a$) // WebRTC VAD; 5–15 s clips $\rightarrow$ 52,020
Step 2: Transcribe
for all clip $c \in \mathcal{C}$ do
  $t_c \leftarrow$ WhisperLargeV3(c, lang=bn)
end for
Step 3: Detect code-switching
for all clip $c \in \mathcal{C}$ do
  Compute n_{bn}, n_{en}, b, and $\rho = n_{\text{en}}/(n_{\text{bn}} + n_{\text{en}})$ from t_c using Bengali and Latin scripts
  $R_c \leftarrow (n_{\text{bn}} + n_{\text{en}}) \geq 4 \wedge n_{\text{bn}} \geq 2 \wedge n_{\text{en}} \geq 1 \wedge b \geq 1 \wedge 0.05 \leq \rho \leq 0.80$
  c.CS $\leftarrow \mathbf{1}[R_c]$
end for
Step 4: Silver-standard annotation
Define NSP(c) as the Whisper large-v3 no-speech probability for clip c
$\mathcal{A} \leftarrow \{c : \text{NSP}(c) < \tau_{\text{lo}}\}$ // auto-accept $\rightarrow$ 39,772
$\mathcal{M} \leftarrow \{c : \tau_{\text{lo}} \leq \text{NSP}(c) < \tau_{\text{hi}}\}$ // borderline band $\rightarrow$ 8,712
for all $c \in \mathcal{M}$ do
  $d \leftarrow$ Qwen2.5Judge(t_c) // 4 yes/no criteria; ACCEPT/REJECT/UNCERTAIN
  if $d =$ ACCEPT then
    $\mathcal{A} \leftarrow \mathcal{A} \cup \{c\}$ // +1,727
  else
    do not add c to $\mathcal{A}$ // REJECT/UNCERTAIN clips conservatively discarded
  end if
end for
Final silver-standard clips: $|\mathcal{A}| = 41{,}499$
Step 5: Normalize transcripts
for all $c \in \mathcal{A}$ do
  $t_c \leftarrow$ Normalize(t_c) // release-time NFC and text cleanup
end for
Step 6: Archive historical split and audit dependence
$(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{dev}}, \mathcal{D}_{\text{test}}) \leftarrow$ HistoricalSplit($\mathcal{A}$) // not source-disjoint; repaired split frozen for future retraining
Historical sizes: 34,605/2,806/4,088 // all reported model scores use this archive
Step 7: Augment training split only
$\mathcal{G}_{\text{train}} \leftarrow$ Augment($\mathcal{D}_{\text{train}}$) // speed $\times$0.9/ $\times$ 1.1; noise (20 dB); pitch $\pm$2 semitones
$\mathcal{D}^{\text{aug}}_{\text{train}} \leftarrow \mathcal{D}_{\text{train}} \cup \mathcal{G}_{\text{train}}$
Training clips after augmentation: 60,936
return $\mathcal{D}$ // released dataset; $\mathcal{D}^{\text{aug}}_{\text{train}}$ used for model training

judge's practical effect, but not its accuracy. The judge has no access to audio and cannot establish acoustic correctness; no superiority over a commercial LLM or manual review is claimed. The two-annotator study in Section III-E instead provides an independent estimate of residual silver-label noise.

Step 5: Transcript Normalization: The stored transcripts receive release-time cleaning, including Unicode NFC normalization, removal of zero-width characters, URLs, hashtags, mentions, and emojis, digit handling, and punctuation and whitespace cleanup. Scoring is a separate process. Reported primary WER/CER and all human-reference results use surface normalization only: they do not apply pronunciation-variant mapping or repetition collapse. WER-RC/CER-RC add only the stated repetition-collapse post-processing. An optional normalization routine maps 650 spelling variants to 271 curated borrowed-word entries, but it is used only for exploratory analysis and supplies none of the reported primary, WER-RC, or human-reference numbers.

The pipeline also includes three supporting lexical resources: about 5,700 known ASR hallucination and noise tokens for transcript filtering, about 18,900 dialectal Bengali word forms protected from standardization, and about 3,700 common English words used as evidence in script-aware token labeling. Their availability does not imply that optional variant mapping was applied to the reported scores.

Step 6: Historical Split and Leakage Audit: All reported predictions and model scores use the archived historical split of 34,605 training, 2,806 development, and 4,088 test clips. A subsequent artifact audit found that this split is not source-disjoint: 69 of the 101 inferred test source IDs also occur elsewhere in the retained corpus, affecting 3,772 test clips, and 326 exact-audio duplicate groups cross the historical test boundary. Missing historical train/dev manifests prevent reconstruction of every training assignment, but the observed overlap is sufficient to rule out a leakage-free generalization claim. We therefore retain the archived scores only as diagnostics. After an additional text-quality filter, the frozen repaired partition contains 29,224/3,855/3,297 original train/dev/test clips, with zero inferred-source or exact-audio overlap across partitions. No system in this paper has been retrained on it. Consequently, the paper does not yet provide a leakage-free model comparison; repaired-split retraining and complete metric regeneration remain required future experimental work.

Step 7: Training Augmentation: Audio augmentation is applied only to the historical training partition. The augmentation methods include speed perturbation ($\times$0.9 and $\times$1.1), additive noise at 20 dB, and pitch shifting by $\pm$2 semitones, which complement spectrogram-level methods such as SpecAugment [40]. Augmented clips keep the same parent identifiers and are not deliberately added to development or test. This prevents augmentation-created copies from being assigned across partitions, but it does not repair the pre-existing source and exact-audio overlap in the historical split. After augmentation, the historical training set contains 60,936 clips.

D. Dataset Statistics

Table III presents the 15 regional labels included in AncholVox, and Fig. 2 shows their clip distribution. The audited construction record contains 52,020 segmented clips (165.23 hours). Language filtering retains 48,772 clips (155.08 hours), and 48,484 of these enter the no-speech/LLM confidence cascade. The final retained set contains 41,499 clips (131.2 hours). Table IV reports its historical train, development, and test partitions.

The main strength of the dataset is its breadth across 15 regional labels and 131.2 hours of retained audio. The

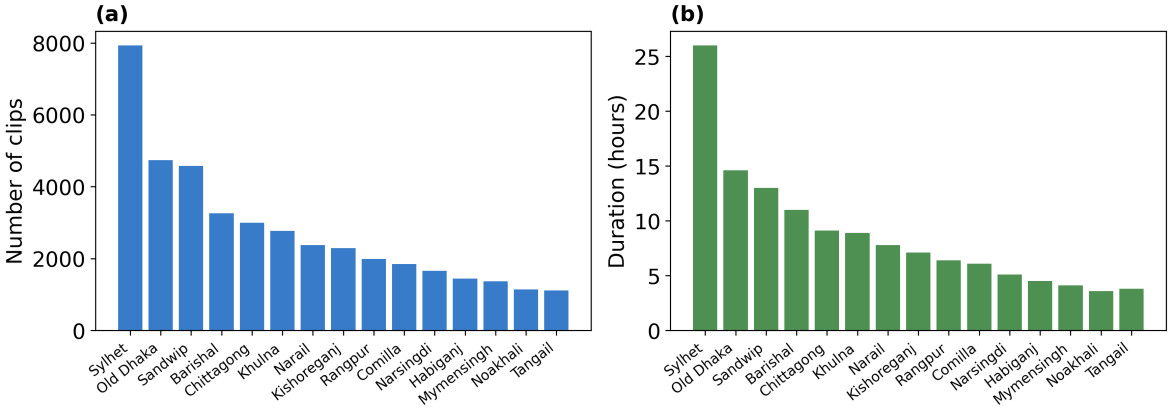


Fig. 2. Dialect distribution across the released dataset: (a) number of clips and (b) total duration in hours.

TABLE III
Dialect coverage in the released dataset.

Dialect	Clips	Hours
Sylhet	7,931	26.0
Old Dhaka	4,741	14.6
Sandwip	4,583	13.0
Barishal	3,259	11.0
Chittagong	2,993	9.1
Khulna	2,774	8.9
Narail	2,371	7.8
Kishoreganj	2,287	7.1
Rangpur	1,985	6.4
Comilla	1,847	6.1
Narsingdi	1,661	5.1
Habiganj	1,446	4.5
Mymensingh	1,369	4.1
Noakhali	1,140	3.6
Tangail	1,112	3.8
Total	41,499	131.2

TABLE IV
Historical train/dev/test counts used by the reported models. Audio augmentation expands the training split to 60,936 clips for the model training, while dev and test remain real and unaugmented. These counts do not describe a verified source-disjoint split; a repaired split is released for future retraining.

Split	Clips	Hours	CS clips	CS%
Train	34,605	108.9	2,019	5.8
Dev	2,806	9.1	166	5.9
Test	4,088	13.2	176	4.3
Total	41,499	131.2	2,361	5.7

script-based rule identifies Bengali–English switching in 5.7% of the silver transcripts. Because this label is derived from automatic text rather than direct acoustic annotation, 5.7% describes the retained transcript distribution, not the verified prevalence of switching in the source speech. CS-specific results are reported on this smaller subset in Section VI.

E. Human Validation of Silver-Standard Labels

As the released transcripts are silver-standard labels, we evaluate them against independent human transcriptions. Two human annotators listened to the same 100 test clips and transcribed the speech independently without access to each other's annotations. Both annotators completed all 100 clips. The silver transcript is treated as the hypothesis and each human transcript as a reference. Every score in this section, in Table V, and in Table XVI is produced by the same primary_normalize routine that defines the primary condition of the silver-reference benchmark: surface cleaning only, with no pronunciation-variant mapping and no repetition collapse. Human-reference and silver-reference numbers are therefore measured under one identical scoring condition.

Against the first annotator, the silver labels obtain a corpus WER/CER of 0.73/0.44; against the second annotator, they obtain 0.80/0.49. These values estimate silver-label noise but should not be interpreted as model accuracy against a fully adjudicated gold reference.

Across all 100 clips, the two human transcripts have a normalized exact-match rate of 7.0%. Using the first annotator as the reference, the second annotator has WER/CER of 0.60/0.38; using the second annotator as the reference, the first annotator has WER/CER of 0.52/0.34. The directional difference is expected because edit-rate denominators depend on the reference length. The low exact-match rate and moderate character-level disagreement reflect both genuine listening differences and legitimate orthographic variation in dialectal Bengali.

The per-dialect results in Table V show substantial variation. The two annotators also differ in which dialects they find difficult, demonstrating that dialectal spelling and listening uncertainty contribute to the measured label noise. Since most dialects have only five to nine clips, these breakdowns are indicative rather than definitive. We use the silver labels as noisy supervision and for relative benchmarking, while the human transcripts provide an independent validation sample rather than an adjudicated gold set. Table VI gives de-identified excerpts from this sample, illustrating both the retained code-switching

TABLE V
Silver-label quality against two independent human references on the 100-clip validation sample. Lower is better; n_{A1} and n_{A2} are the numbers of completed transcripts for Annotator 1 and Annotator 2.

Dialect	n_{A1}	WER_{A1}	CER_{A1}	n_{A2}	WER_{A2}	CER_{A2}
Barishal	8	0.73	0.45	8	0.73	0.43
Chittagong	7	0.89	0.53	7	0.76	0.44
Comilla	5	0.94	0.48	5	0.86	0.47
Habiganj	7	1.00	0.66	7	0.89	0.56
Khulna	6	0.68	0.32	6	0.76	0.45
Kishoreganj	8	0.58	0.32	8	0.79	0.42
Mymensingh	9	0.84	0.49	9	0.84	0.43
Narail	6	0.71	0.31	6	0.67	0.29
Narsingdi	6	0.76	0.43	6	0.81	0.44
Noakhali	5	1.09	0.64	5	0.76	0.42
Old Dhaka	8	0.60	0.37	8	0.86	0.70
Rangpur	8	0.75	0.54	8	0.83	0.63
Sandwip	5	0.38	0.35	5	0.85	0.68
Sylhet	6	0.43	0.31	6	0.88	0.72
Tangail	6	0.76	0.47	6	0.75	0.41
Overall	100	0.73	0.44	100	0.80	0.49

spans and a clear silver-label failure.

F. Ethical and Legal Considerations

The source material is publicly accessible social-media content, but public accessibility alone does not establish permission to redistribute audio or eliminate creators' privacy interests. We therefore retain source identifiers and timestamps, omit speaker names, handles, faces, contact information, and demographic labels, and restrict the intended use to non-commercial ASR research. The corpus contains 1,007 source records: 36,967 clips in the general collection and 4,532 in the UK-diaspora collection. These counts describe collection domains, not population strata.

No consented age, gender, socioeconomic, education, or urban–rural metadata are available, and we do not infer sensitive attributes from voices or online profiles. Consequently, AncholVox cannot support demographic subgroup-fairness claims. Platform visibility, search-query design, and internet access may overrepresent digitally connected, urban, younger, or diaspora creators; these characteristics are plausible selection mechanisms, not measured labels. Dialect coverage therefore reflects online availability rather than population prevalence. Fig. 2 reports the observable dialect imbalance so that users can apply weighting or resampling, while future consent-based collection should record self-reported demographic variables for a proper fairness audit.

The release follows a rights-aware governance procedure. It provides a documented takedown and opt-out channel, prohibits speaker identification, surveillance, and re-identification, and records provenance at the source level.

We state the current clearance status explicitly, because describing a governance procedure is not the same as having completed it. Three determinations are treated as blocking for any public redistribution of audio: institutional confirmation of whether formal ethics review is required, legal or institutional review of the redistribution basis for each source platform and content category, and a recorded source-level allow/remove decision. At the time of writing these three determinations are pending, and they are tracked with named owners and dated evidence in the ETHICS_AND_DATA_GOVERNANCE.md register released with the code.

Neither the public accessibility of the source material nor the non-commercial licence we attach to the compilation authorizes redistribution of third-party social-media audio; only those determinations can. Until they are recorded, the audio release is provisional and the intended distribution is the metadata-only fallback: source pointers, timestamps, clip-level split manifests with checksums, transcripts where permitted, and reproducible processing code, without hosted audio. If review later declines a source, that source is removed under a versioned removal log so mirrors can honour the decision.

These safeguards do not remove all risk: speech is potentially identifying, creators did not provide study-specific consent, and models may perform unevenly across dialects. We therefore report these limitations explicitly, provide per-dialect results, and maintain a removal process throughout the lifetime of the resource.

IV. ASR Models and Fine-Tuning

This work evaluates three ASR model families. These models are selected to represent the two main ASR approaches and different types of pre-training.

Whisper (Sequence-to-Sequence): Whisper [19] is an encoder-decoder transformer model trained on 680k hours of weakly supervised multilingual audio. It is important for this work because its subword vocabulary includes both Bengali and Latin-script tokens. This allows Whisper to generate mixed-script, code-switched output in a single decoding step, making it well suited for Bengali English code-switching. Although convolution-augmented models such as Conformer [41] are also strong for ASR, Whisper is especially useful here because of its shared multilingual vocabulary. For fine-tuning, we use Whisper-small, which has 244M parameters. This model is practical to train on modest hardware while still benefiting from large-scale pre-training. During inference, Bengali transcription is forced for all Whisper models.

Meta MMS (CTC, Multilingual): MMS [20] is a CTC-based multilingual ASR model that supports more than 1,000 languages through language-specific adapters. We include it as a strong multilingual zero-shot baseline. However, its Bengali output vocabulary is monolingual, so it cannot generate English-script words. This makes MMS useful for comparison, because it helps show how much code-switching affects models that are not designed to produce mixed-script output.

wav2vec2-XLS-R (CTC, Self-Supervised): wav2vec2-XLS-R-300M [17] is a self-supervised multilingual speech model pre-trained on 436k hours of audio. We use a Bengali fine-tuned checkpoint as a zero-shot baseline. We also fine-tune the model on AncholVox so that it can be compared with Whisper using the same training data.

TABLE VI
De-identified qualitative examples from the existing 100-clip, two-annotator validation sample. Excerpts are shortened and lightly romanized for compact presentation; they are illustrations rather than new annotations.

ID	Dialect	Silver excerpt	Independent-reference excerpt	What it illustrates
E1	Rangpur	"assalamu alaikum to everyone guys, this is your boy …welcome to my channel"	"assalamu alaikum to everyone guys, this is your boy …welcome to my channel"	Long mixed-language span is retained, with minor Bengali spelling variation.
E2	Comilla	"I think Mainamati museum-e …What's inside? Forest?"	"I think Mainamati museum …What's inside? Forest?"	Natural intra-utterance switching and an English phrase embedded in Bengali.
E3	Barishal	"eta change hoye jabe; the day after tomorrow …the day before"	"eta change hoiya jabe; the day after tomorrow …the day before"	English temporal phrases are preserved despite dialectal spelling variation.
E4	Narsingdi	"emijmen prok ek grem holade prok"	"this is an amusement park, a green holiday park"	Clear silver-label failure, demonstrating why the corpus is not a gold standard.

This model represents the CTC-based ASR approach in our experiments.

Together, these systems cover multilingual encoder–decoder and CTC paradigms, monolingual versus mixed-script output vocabularies, and zero-shot versus in-domain adaptation. They do not constitute an exhaustive leaderboard. SeamlessM4T is Bengali-capable and would be a useful future comparison, whereas the released Canary-1B vocabulary does not include Bengali [38], [39]. We therefore restrict conclusions to the evaluated model set.

Fine-tuning Setup: Whisper is fine-tuned using the Hugging Face Transformers [42] and Datasets [43] libraries with the Adam optimizer [44]. We use a learning rate of 10^{-5}, an effective batch size of 16, three training epochs, and 500 warm-up steps. The best checkpoint is selected based on development WER. For the CTC model, we build a character vocabulary from the training transcripts, covering Bengali, Latin characters, and common punctuation. The CTC output head is reinitialized, and the encoder is fine-tuned for three epochs with a learning rate of 10^{-4}. All fine-tuning and benchmark inference, including both zero-shot and fine-tuned systems, are run on a single NVIDIA A100-SXM4-40GB GPU. The one-time silver-standard transcription of all clips using Whisper large-v3 is performed on a CPU workstation with an Intel Core i9-14900 processor, 24 cores, and 32 GB RAM.

The CTC model is trained by minimizing the negative log-likelihood over all possible alignments π that collapse to the target sequence y:

$$\mathcal{L}_{\text{CTC}} = -\log \sum_{\pi \in \mathcal{B}^{-1}(y)} \prod_{t} p(\pi_t \mid x) \tag{2}$$

where $\mathcal{B}$ is the CTC collapsing function. In contrast, Whisper is trained using the standard token-level cross-entropy loss of its decoder. The main hyperparameters are reported in Table VII. Fig. 3 shows that all Whisper variants converge within about 2k steps. The CS_ONLY configuration runs for fewer steps because it uses only the approximately 4.5k code-switched clips.

Comparative Training Configurations: We train three Whisper variants: BN_ONLY, which uses only clips whose silver transcript is Bengali-only; CS_ONLY, which uses only clips whose silver transcript is code-switched; and ALL, which uses both. The base model, optimizer settings, and historical test set are shared, but the configurations have substantially different sample sizes, lexical coverage, and numbers of optimization steps. Their results therefore compare practical training recipes; they do not isolate the causal effect of training-data composition.

TABLE VII
Fine-tuning hyperparameters for the two trained model families.

Hyperparameter	Whisper-small	wav2vec2-XLS-R
Base parameters	244M	300M
Architecture	seq2seq	CTC
Learning rate	10^{-5}	10^{-4}
Effective batch size	16	16
Epochs	3	3
Warm-up steps	500	500
Precision	bf16/fp16	bf16/fp16
Model selection	best dev WER	best dev WER

V. Experimental Setup

This section defines the evaluation metrics, scoring conditions, baseline systems, and statistical tests used for dialectal and code-switched Bengali speech.

A. Evaluation Metrics

We use Word Error Rate (WER) and Character Error Rate (CER) as the main evaluation metrics. Because the benchmark references are automatic silver labels, these scores measure surface agreement with the silver reference, not definitive transcription accuracy. CER is especially important in this work because dialectal Bengali does not have a standardized spelling system. As a result, WER may penalize valid transcriptions that use different spellings for the same spoken word, while CER measures character-level agreement with the same noisy reference.

To characterize code-switched speech, we report WER separately for Bengali-script and English-script reference words within the 176 silver-labelled code-switched clips

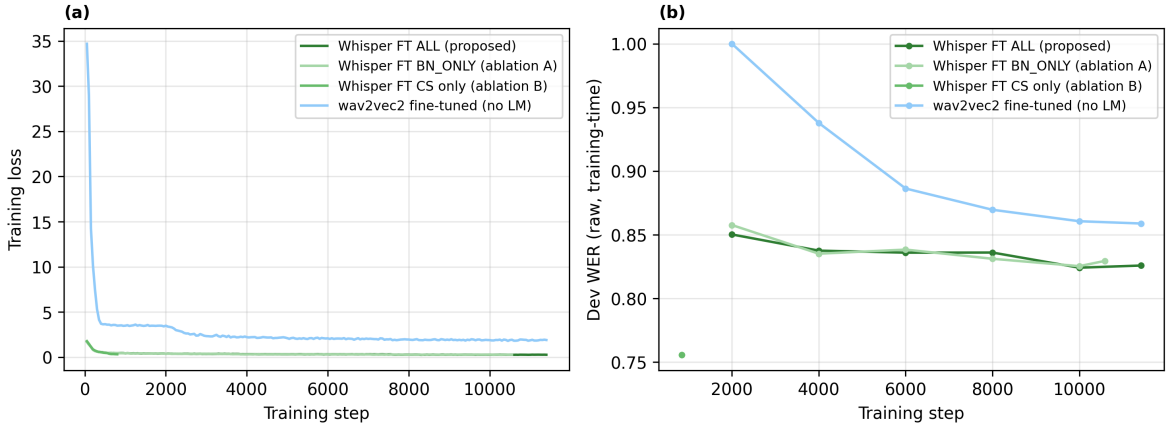


Fig. 3. (a) Training loss and (b) development WER during fine-tuning. Development WER follows the training-time evaluation pipeline and is not directly comparable to the separately reported test WER-RC condition.

(WER-BN-CS and WER-EN-CS). These are not scores for Bengali-only or English-only clips. Both are computed on the same code-switched subset, so they share a scope with the language-balanced WER (LB-WER) defined as the unweighted mean

$$\text{LB-WER} = \tfrac{1}{2}(\text{WER-BN-CS} + \text{WER-EN-CS}). \quad (3)$$

Keeping all three quantities on one subset is what makes the identity above hold in the reported tables; computing the components corpus-wide and the mean on the CS subset would combine incompatible scopes. LB-WER gives the two scripts equal influence despite unequal token counts; it does not test a joint word-and-language label.

Code-Switch Detection F1 (CS-F1) measures whether the hypothesis contains both Bengali and English according to the script-based detector. Unlike the metrics above, CS-F1 is a clip-level binary classification score evaluated over the entire test set: all 4,088 clips are classified, of which 176 are code-switched positives and 3,912 are negatives. Precision therefore depends on false positives raised on Bengali-only clips, which is the intended behaviour — a system that emits spurious Latin tokens everywhere should be penalized. CS-F1 is thus not restricted to the 176-clip subset, even though its recall term and the confidence intervals we report are governed by that small positive count.

We distinguish two scoring conditions. The primary condition applies conventional surface normalization (Unicode normalization, case handling, punctuation removal, and whitespace cleanup) while retaining every repeated character and word, so decoder insertion errors remain in WER and CER. The WER-RC/CER-RC condition additionally collapses excessive character and consecutive-word repetitions and is reported as explicit system post-processing. All models use the same procedure within each condition.

Formally, WER is calculated using substitutions, deletions, and insertions:

$$\text{WER} = \frac{S + D + I}{N} \quad (4)$$

where S, D, and I represent substitution, deletion, and insertion errors, and N is the total number of words. WER can exceed 1.0 when insertion errors are large; in this work, that happens on a small number of hallucination-heavy zero-shot outputs, so values above 1.0 are expected and do not indicate a scoring bug.

CER is calculated in the same way but at the character level:

$$\text{CER} = \frac{S_c + D_c + I_c}{N_c} \quad (5)$$

CER complements WER because dialectal spellings may differ even when their character sequences remain similar; it is not treated as a replacement for human-reference evaluation.

For code-switch detection, we treat it as a yes/no problem: whether code-switching appears in the output or not. We use Precision, Recall, and F1 score:

$$\text{P} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{R} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{CS-F1} = \frac{2PR}{P + R} \quad (6)$$

Finally, English-word survival (EN-Surv) measures how many English words in the reference are correctly preserved in the model output for code-switched speech:

$$\text{EN-Surv} = \frac{\sum_{(r,h)\in\mathcal{P}} |\{w \in r : \text{English}(w)\} \cap h|}{\sum_{(r,h)\in\mathcal{P}} |\{w \in r : \text{English}(w)\}|} \quad (7)$$

where $\mathcal{P}$ is the set of code-switched (reference, hypothesis) pairs.

In the discussion below, EN-Surv is our primary code-switching metric because it directly measures English-token retention. WER-CS, WER-BN-CS, WER-EN-CS and LB-WER are reported as supporting diagnostics computed on the smaller 176-clip code-switched test subset; CS-F1 is a detection score over the whole 4,088-clip test set, and only its 176 positives limit the precision of its recall term.

B. Baselines

We compare our method with four zero-shot baseline models. These include Whisper-small (used as a base model), Whisper large-v3 (the largest evaluated Whisper checkpoint), MMS-1B, and wav2vec2-XLS-R-BN, which is trained on standard Bengali speech.

Whisper large-v3 also generated the silver labels, so its 0.930 evaluation WER requires an artifact-level explanation. The two relevant comparisons are distinct. First, treating each archived construction-time transcript as the reference and the paired evaluation-time large-v3 output as the hypothesis gives 0.927 WER. Second, comparing the same evaluation-time outputs with the stored silver references gives 0.930 WER. By contrast, construction-time transcripts and stored silver references differ by only 0.005 WER after final text cleaning.

The current source paths are materially different: construction defaults to faster-whisper/CTranslate2 large-v3 with beam size 1, condition_on_previous_text=False, word timestamps, and no VAD; benchmark inference uses Hugging Face Transformers, batches of 16, forced Bengali transcription, and a 128-new-token cap. Current scripts record package versions, requested and resolved revisions where available, decoding metadata, and audio SHA-256 values on new runs. The archived rows contain none of those fields, however, so neither historical model revision nor historical software versions can be recovered. Present local WAV hashes likewise cannot prove the bytes used in remote historical inference. Artifact hashes and the precise known/unknown provenance fields are released in REFERENCE_PIPELINE_PROVENANCE.md. The discrepancy therefore remains unresolved and is not treated as a self-reproduction check.

C. Significance Testing

To quantify uncertainty, we use two-sided paired bootstrap tests [45] with 10,000 clip-level resamples. For each resample, corpus WER is recomputed from the summed edit and reference-word counts. The two-sided p-value is twice the smaller bootstrap tail probability around zero, with a plus-one Monte Carlo correction; percentile intervals provide 95% confidence intervals. Tests use a fixed seed and are reported separately for primary WER and WER-RC. Clips cut from the same or related source videos are not statistically independent, so clip resampling can understate uncertainty. The released analysis also reports exploratory source-video-cluster resampling, but inferred source IDs do not establish speaker independence and the contaminated historical split remains unresolved. We consequently treat all significance tests involving fine-tuned systems as diagnostic rather than as generalization evidence.

VI. Results

Table VIII and Fig. 4 compare zero-shot and task-adapted systems under both primary and repetition-collapsed scoring.

The zero-shot models show limited silver-reference agreement: Whisper large-v3 obtains 0.930 primary WER and Whisper-small obtains 1.142. On the historical split, joint-data Whisper scores 0.964 and fine-tuned wav2vec2 has the lowest primary WER, 0.861. These differences are diagnostic associations with the training recipes and cannot establish leakage-free adaptation gains.

The Whisper training configurations expose a trade-off. BN_ONLY has lower primary WER than ALL (0.923 versus 0.964) but almost no code-switch detection (0.011 CS-F1). CS_ONLY has the strongest English and switching indicators (0.737 WER-EN-CS and 0.921 LB-WER on CS clips; 0.276 CS-F1) but the weakest fine-tuned primary WER (1.065). ALL occupies the middle ground. Its primary WER/CER are 0.964/0.597, while explicit repetition collapse gives 0.801/0.490 WER-RC/CER-RC. The WER-RC score differs from all zero-shot systems and fine-tuned wav2vec2 ($p = 0.0002$), but not from BN_ONLY ($p = 0.911$); its clearer descriptive difference from BN_ONLY is in the switching measures.

A. Per-Dialect Results

Table IX reports the repetition-collapsed Whisper ALL scores, while Fig. 5 and Table X retain repetitions and therefore show primary per-dialect WER. Both conditions reveal substantial dialect-to-dialect variability.

Table X compares representative models under primary scoring. Fine-tuned wav2vec2 has the lowest overall WER and leads on six dialects; Whisper ALL leads on six, and large-v3 leads on three. This heterogeneous ranking motivates per-dialect reporting rather than a single universal claim.

VII. Analysis and Discussion

This section analyzes model behavior beyond overall error rates. It examines error patterns, code-switching behavior, and qualitative failure cases across different models and dialects. It also evaluates how well models preserve code-switching structure, compares model outputs with human references, and reports statistical significance. The goal is to characterize the observed differences and identify which parts of dialectal and code-switched ASR remain difficult.

A. Error Types and English-Word Survival

Table XI separates WER into three error types: substitutions (S), deletions (D), and insertions (I). It also reports English-word survival (EN-Surv), which measures the proportion of English words in the reference that are correctly preserved in the model output for code-switched clips. Fig. 6 shows the substitution, deletion, and insertion breakdown visually.

A clear pattern is observed. Monolingual CTC models and the BN_ONLY configuration preserve almost no exact English tokens. CS_ONLY reaches the highest EN-Surv among fine-tuned models (0.37), while ALL reaches 0.13. Large-v3 preserves the most English tokens overall (0.65)

TABLE VIII
Agreement with the AncholVox silver references on the historical test set. WER, WER-RC and CER are corpus-wide over all 4,088 test clips; WER retains all repetitions, while WER-RC applies explicit repetition-collapse post-processing. WER-BN-CS, WER-EN-CS, LB-WER and WER-CS are all scoped to the same 176 code-switched test clips, so LB-WER is exactly the equal-weight mean of the WER-BN-CS and WER-EN-CS columns in the same row. CS-F1 is a clip-level detection score over all 4,088 clips (176 positives). Bold numeric values are column minima (or the maximum for CS-F1); the bold model name identifies the joint-data Whisper system.

Model	WER	WER-RC	CER	WER-BN-CS	WER-EN-CS	LB-WER	WER-CS	CS-F1
Zero-shot baselines								
Whisper-small	1.142	1.066	1.031	1.000	1.463	1.232	1.133	0.000
Whisper large-v3	0.930	0.910	0.757	0.818	1.106	0.962	0.818	0.144
MMS-1B	0.989	0.990	0.571	1.000	1.131	1.066	0.940	0.041
wav2vec2-XLS-R-BN	0.963	0.963	0.622	0.964	1.039	1.001	0.935	0.044
CTC fine-tuned on AncholVox								
wav2vec2	0.861	0.861	0.493	0.942	0.996	0.969	0.877	0.135
Whisper comparative training configurations								
BN_ONLY (A)	0.923	0.800	0.575	1.047	0.994	1.021	0.933	0.011
CS_ONLY (B)	1.065	0.883	0.679	1.105	0.737	0.921	0.963	0.276
ALL (joint data)	0.964	0.801	0.597	1.106	0.902	1.004	0.959	0.192

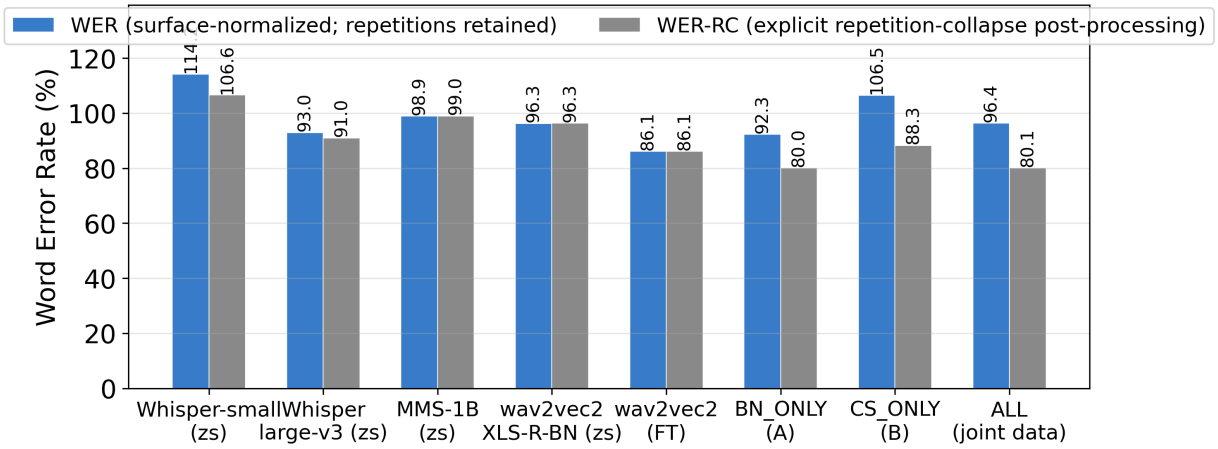


Fig. 4. Surface-normalized WER (repetitions retained) and WER-RC after explicit repetition-collapse post-processing. The large Whisper-specific gap shows why the two conditions must be reported separately.

and obtains 0.930 primary WER, whereas ALL obtains 0.964 primary WER but 0.801 WER-RC. The joint-data recipe is associated with higher switching measures than BN_ONLY, while decoder repetition remains a separate weakness.

B. Code-Switching Error Analysis

We first clarify the scope of the code-switching results. The per-clip CS error rates — WER-CS, WER-BN-CS, WER-EN-CS and LB-WER — are computed on only 176 code-switched test clips. WER-BN-CS means Bengali-script words within those mixed clips; it is distinct from whole-clip WER on the 3,912 Bengali-only clips, denoted WER-BN-only below. CS-F1 is scoped differently: every one of the 4,088 test clips is classified as code-switched or not, so its false positives come from the Bengali-only clips even though only 176 positives are available to estimate recall. Therefore, these results should be treated as indicative rather than highly precise. The strongest descriptive pattern is that monolingual and Bengali-only models almost completely drop exact English tokens, but this too requires confirmation against adjudicated references.

Because only 176 test clips are labeled as code-switched, whole-clip WER-CS and the recall side of CS-F1 are treated as descriptive supporting measures rather than definitive rankings. English-word survival provides a directly interpretable complementary measure, but it likewise requires confirmation on a larger adjudicated subset.

Bootstrap intervals from the existing predictions quantify this uncertainty. For Whisper ALL, WER-CS is 0.959 (95% CI $[0.868, 1.063]$), CS-F1 is 0.192 ($[0.120, 0.264]$), and English-word survival is 0.127 ($[0.064, 0.201]$). CS_ONLY has the higher CS-F1, 0.276 ($[0.220, 0.328]$). The paired

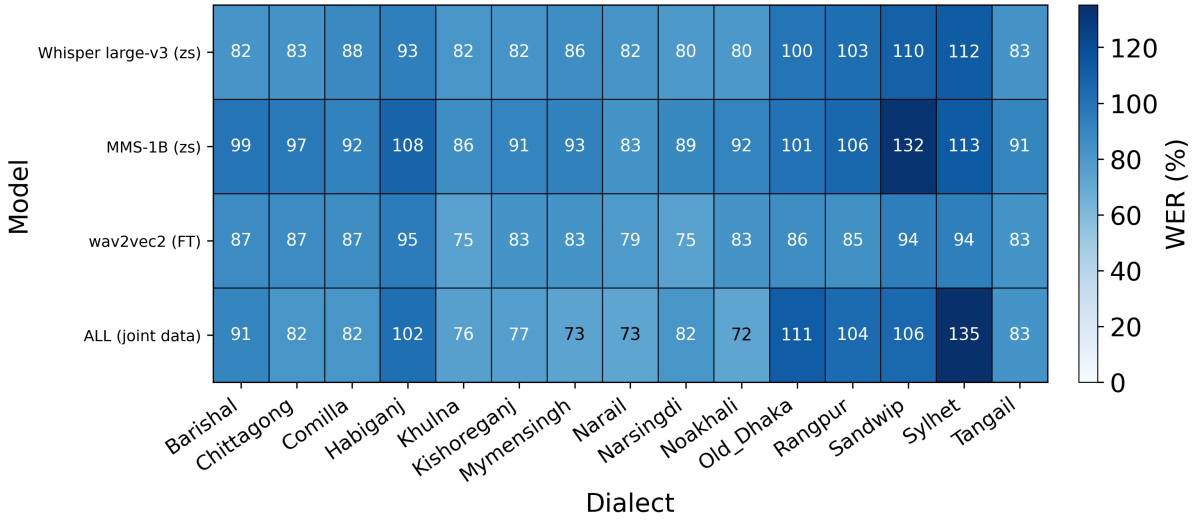


Fig. 5. Primary surface-normalized per-dialect WER for four representative systems; repetitions are retained and darker cells indicate higher WER. Complete eight-model results remain available in the released metrics.

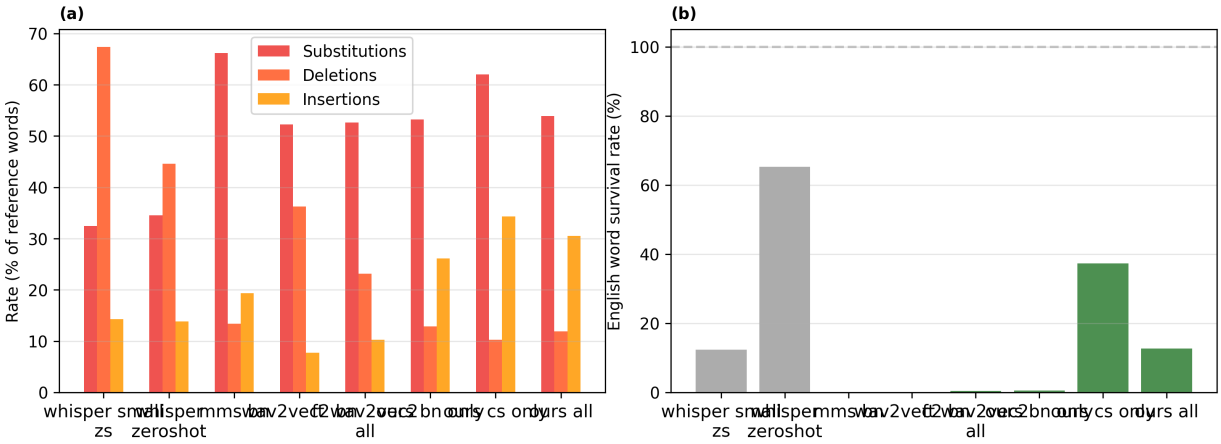


Fig. 6. Error analysis by model: (a) substitution, deletion, and insertion rates per reference word; and (b) English-word survival rate on code-switched clips.

CS-F1 difference for ALL minus BN_ONLY is +0.181 ($[0.111, 0.253]$), whereas ALL minus CS_ONLY is −0.083 ($[-0.153, -0.015]$). Thus, on these historical predictions, ALL has higher CS-F1 than BN_ONLY but lower CS-F1 than CS_ONLY; this is a descriptive recipe comparison rather than a causal training effect.

Table XII gives a descriptive comparison of primary WER on code-switched and Bengali-only clips, and Fig. 7 shows the same contrast alongside the detection scores. Most gaps are small or negative, so this comparison does not support a causal claim that the CS subset is intrinsically harder; the subsets differ in content and length. The clearer finding is token-specific: English survival is near zero for monolingual and BN_ONLY systems, while CS-supervised Whisper systems recover more mixed-script output.

Beyond checking whether English words are preserved, we compare the number of language transitions produced by each system. Walking through a transcript from left to right, we label every language-bearing token as Bengali or English and count a switch at each adjacent pair whose labels differ; the 176 code-switched test clips contain 463 reference switches. The count-recall diagnostic is $\sum_c \min(\hat{s}_c, s_c) / \sum_c s_c$, where s_c and $\hat{s}_c$ are the reference and hypothesis switch counts of clip c; the cap limits rewards for over-generation. This statistic compares counts only: it does not determine whether a switch occurred at the correct word or boundary, so it is not a positional-preservation score. We report it only for the Whisper sequence-to-sequence models, because they can naturally generate mixed Bengali and Latin script. Monolingual CTC models such as MMS are excluded because they cannot reliably produce Latin-script words, with EN-survival of 0.00 in Table XI. To reduce character-level artifacts, a Latin token is counted as English only if it contains at least two letters.

Table XIII reports this count-based analysis. Reference code-switched clips contain an average of 2.63 language

TABLE IX
Per-dialect WER-RC/CER-RC for Whisper ALL after explicit repetition-collapse post-processing.

Dialect	Clips	WER-RC	CER-RC
Sylhet	1,111	0.93	0.65
Old Dhaka	360	0.78	0.53
Sandwip	176	0.92	0.67
Barishal	282	0.87	0.54
Chittagong	198	0.79	0.46
Khulna	120	0.66	0.39
Kishoreganj	743	0.75	0.36
Comilla	157	0.78	0.42
Narail	464	0.70	0.41
Rangpur	165	0.79	0.51
Narsingdi	60	0.72	0.39
Mymensingh	56	0.70	0.35
Habiganj	75	0.90	0.52
Tangail	51	0.74	0.39
Noakhali	70	0.70	0.36
Overall	4,088	0.80	0.49

TABLE X
Primary per-dialect WER for representative models; repetitions are retained. Bold = best per dialect.

Dialect	large-v3	wav2vec2	Whisper ALL
Sylhet	1.124	0.936	1.352
Old Dhaka	0.997	0.861	1.114
Sandwip	1.096	0.941	1.057
Barishal	0.821	0.872	0.907
Chittagong	0.829	0.872	0.819
Khulna	0.817	0.753	0.756
Kishoreganj	0.822	0.832	0.770
Comilla	0.883	0.866	0.817
Narail	0.822	0.792	0.728
Rangpur	1.028	0.850	1.041
Narsingdi	0.799	0.750	0.817
Mymensingh	0.863	0.829	0.734
Habiganj	0.935	0.951	1.021
Tangail	0.827	0.831	0.831
Noakhali	0.803	0.825	0.724
Overall	0.930	0.861	0.964

switches, and every Whisper system produces fewer. Within the fine-tuned family, preservation tracks the amount of code-switch supervision exactly as expected: CS_ONLY preserves 36% of reference switches, ALL 12%, and BN_ONLY almost none (2%). Zero-shot large-v3 is highest overall (46%), consistent with it also having the highest English survival in Table XI. A model can match the count while placing switches at the wrong words, and the measure may still be inflated by over-generation that the cap does not fully remove. It is therefore reported as a descriptive count-recall upper bound, not evidence that the correct switching positions were preserved. Relative to BN_ONLY, the joint-data recipe is associated with higher code-switch detection (Table VIII); relative to CS_ONLY, however, it has lower switch-count recall.

C. Qualitative Error Analysis

Inspection of model outputs shows three repeated error patterns. For readability, the examples below use romanized Bengali and omit Bengali script.

TABLE XI
Primary-score error rates per reference word and English-word survival (EN-Surv); repetitions are retained.

Model	S	D	I	EN-Surv
Whisper large-v3 (zs)	0.35	0.45	0.14	0.65
Whisper-small (zs)	0.32	0.67	0.14	0.12
MMS-1B	0.66	0.13	0.19	0.00
wav2vec2 (zs)	0.52	0.36	0.08	0.00
wav2vec2 (FT)	0.53	0.23	0.10	0.00
Whisper BN_ONLY	0.53	0.13	0.26	0.01
Whisper CS_ONLY	0.62	0.10	0.34	0.37
Whisper ALL	0.54	0.12	0.31	0.13

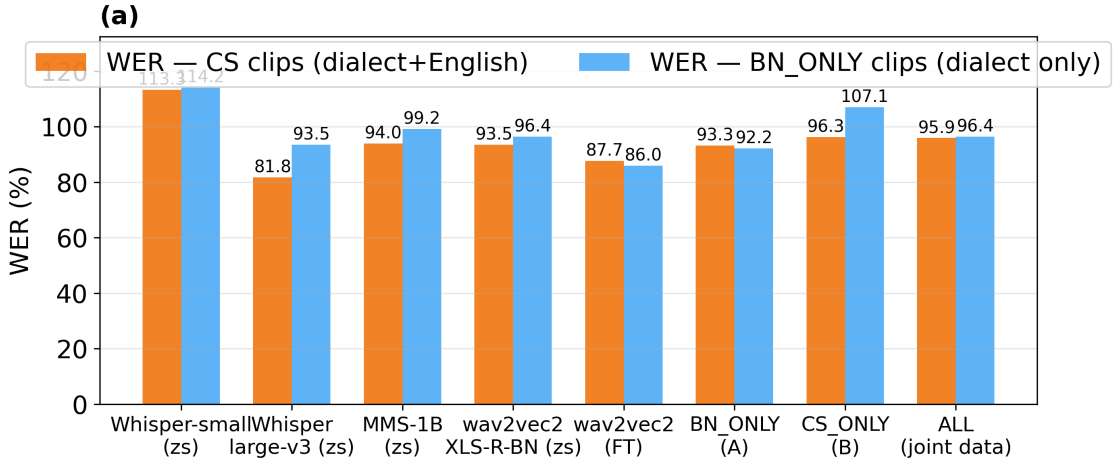
TABLE XII
Primary whole-clip WER on CS vs. Bengali-only clips. Gap = WER-CS − WER-BN-only. The bold model name identifies the joint-data system.

Model	WER-CS	WER-BN-only	Gap
Whisper large-v3 (zs)	0.818	0.936	−0.118
Whisper-small (zs)	1.133	1.142	−0.009
MMS-1B	0.94	0.99	−0.05
wav2vec2 (FT)	0.877	0.860	+0.017
Whisper CS_ONLY	0.963	1.071	−0.108
Whisper ALL	0.959	0.964	−0.005

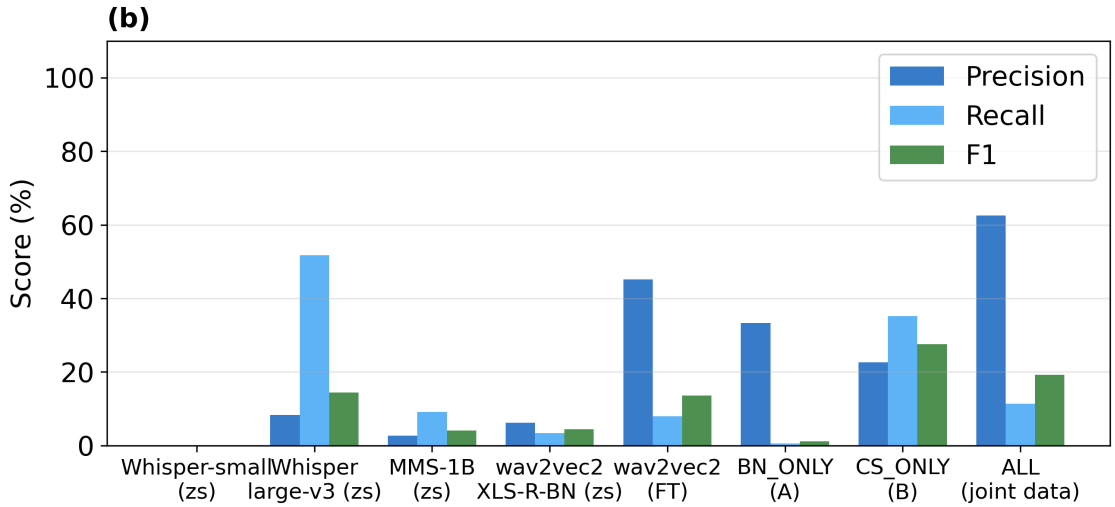
- Repetition hallucination occurs when Whisper generates repeated words on difficult clips. For example, a short reference such as “tai” may be expanded into many repeated tokens. Primary WER counts every insertion; WER-RC reports the behavior of an explicitly post-processed system.
- English deletion occurs when monolingual or Bengali-focused models remove English words from code-switched speech. For example, the reference “ami office-e meeting korlam” may be decoded as “ami … korlam”, where the English words are missing. This explains the near-zero English-word survival shown in Table XI.
- Spelling variation occurs when fine-tuned models produce phonetically correct words using different spellings, such as “kortasi” and “kortechi” for the same spoken form. These cases are counted as word-level errors, but they do not always indicate real recognition failure. This also explains why CER is much lower than WER.

These patterns explain the main trends in the results. Fine-tuning mainly reduces hallucination and English-word deletion, while many remaining errors come from acceptable spelling differences caused by the lack of standardized dialect spelling.

Table XIV expresses substitutions, deletions, and insertions as shares of all primary word errors. Large-v3 has a 48% deletion share. For ALL, the deletion share falls to 12%, but the insertion share rises to 32%, exposing the repetition behavior hidden by WER-RC. Approximately 14–15% of errors in the fine-tuned systems satisfy the near-spelling threshold, against 5% for zero-shot large-v3, so fine-tuning shifts error mass from outright deletion toward orthographic near-misses. These are reported as potential orthographic variants, not automatically rela-



(a) WER on code-switched and Bengali-only clips.



(b) Code-switch detection precision, recall, and F1.

Fig. 7. Code-switching analysis: (a) surface-normalized WER on code-switched and Bengali-only clips; and (b) code-switch detection precision, recall, and F1. Models never trained on CS show near-zero CS-F1.

TABLE XIII
Code-switch counts on the 176 CS test clips (Whisper seq2seq family): average intra-utterance language switches produced and capped count recall over the 463 reference switches, $\sum_c \min(\hat{s}_c, s_c)/\sum_c s_c$. Reference clips average 2.63 switches. This does not evaluate exact boundary positions. Computed by evaluation/17_cs_structure_preservation.py.

Model	Sw./clip	Count recall
Reference	2.63	—
Whisper large-v3 (zs)	1.50	46%
Whisper CS_ONLY	1.35	36%
Whisper ALL	0.31	12%
Whisper BN_ONLY	0.05	2%

TABLE XIV
Error taxonomy showing deletion, insertion, and substitution shares, plus benign spelling variants, each as a share of all word errors ($S + D + I$) recovered from the jiwer alignment. A spelling variant is a substitution whose normalized character edit distance $\text{lev}(r, h)/\max(|r|, |h|)$ is ≤ 0.34. Computed by evaluation/18_spelling_variant_analysis.py; such pairs are reported, never relabeled as correct.

Model	Del	Ins	Sub	Spell. var.
Whisper large-v3 (zs)	48%	15%	37%	5%
wav2vec2 (FT)	27%	12%	61%	15%
Whisper ALL	12%	32%	56%	14%

beled as correct.

D. Training-Recipe and Post-Processing Comparisons

BN_ONLY, CS_ONLY, and ALL compare recipes with unequal data budgets; they do not isolate the causal effect of the training subset. Descriptively, BN_ONLY has stronger primary WER than ALL but almost no code-switch detection, while CS_ONLY has stronger English retention but substantially weaker overall WER. Repetition collapse reduces ALL from 0.964 to 0.801 WER and from 0.597 to 0.490 CER without changing model parameters. Because this operation removes genuine insertion errors, the uncollapsed result remains primary and the collapsed result is labeled WER-RC/CER-RC throughout.

TABLE XV
Explicit system post-processing comparison for Whisper ALL. Primary WER/CER retain repetitions; WER-RC/CER-RC collapse degenerate runs.

Configuration	WER	CER
Primary: repetitions retained	0.964	0.597
System output: repetition collapse	0.801	0.490

E. Case Study

A single Sylheti test clip illustrates decoder repetition: a one-word reference is expanded by zero-shot Whisper-small into roughly forty repeated tokens. Primary scoring correctly counts these as insertions. Repetition-collapse post-processing shortens such degenerate runs, explaining the WER-RC change in Table XV; it does not constitute an improvement to the underlying acoustic model. A second recurring failure is English-token deletion by monolingual and Bengali-only systems.

F. Statistical Significance

Because the historical split is contaminated and clips from related source videos are dependent, the following clip-bootstrap results are secondary descriptive evidence, not confirmatory tests of generalization. Exploratory source-video-cluster resampling is included in the released report, but it cannot undo train–test source overlap or establish speaker independence.

Under that limited interpretation, the primary-WER bootstrap reports lower ALL error than Whisper-small ($p = 0.0002$) and MMS-1B ($p = 0.0066$), no detectable difference from the zero-shot wav2vec2 Bengali checkpoint ($p = 0.9411$), and higher error than large-v3 ($p = 0.0012$), fine-tuned wav2vec2 ($p = 0.0002$), and BN_ONLY ($p = 0.0002$). For WER-RC, ALL has a 95% interval of $[0.789, 0.813]$; its clip-bootstrap difference from BN_ONLY is not detectable ($p = 0.9109$). Code-switch intervals are wider because the test set contains only 176 transcript-labeled positive clips. The estimates therefore describe the archived artifacts and do not establish a universal model ranking.

G. Interpreting High Error Rates

Primary WER remains high for every system, ranging from 0.861 to 1.142. Values above 1.0 arise when insertions exceed the number of reference words. This reflects difficult dialectal speech, non-standard spelling, decoder failures, and noise in Whisper-derived silver references. CER and WER-RC illuminate specific aspects of that difficulty, but neither removes the need for primary WER and human-reference evaluation. AncholVox is therefore positioned as a stress-test benchmark with transparent scoring conditions.

H. Human-Reference Evaluation

Since the test labels are silver-standard, the main benchmark may favor systems whose output resembles Whisper.

TABLE XVI
Model errors on the 100-clip validation sample using the two independent human references. Both annotators completed all 100 references. Scoring uses the same primary condition as Table VIII (surface normalization only; no pronunciation-variant mapping and no repetition collapse), so the two tables are directly comparable. Lower values are better.

Model	WER_{A1}	CER_{A1}	WER_{A2}	CER_{A2}
MMS-1B (zero-shot)	0.76	0.39	0.70	0.36
wav2vec2 (fine-tuned)	0.85	0.47	0.82	0.47
wav2vec2-XLS-R-BN (zero-shot)	0.87	0.54	0.86	0.57
Whisper large-v3 (zero-shot)	0.92	0.74	0.86	0.71
Whisper BN_ONLY	0.92	0.56	0.86	0.52
Whisper ALL (joint data)	0.93	0.54	0.86	0.50
Whisper CS_ONLY	0.93	0.56	0.86	0.51
Whisper-small (zero-shot)	1.11	1.00	1.03	0.96

We therefore evaluate every model against both human transcripts in the 100-clip validation sample, under the same primary scoring condition used for Table VIII. The ranking changes substantially. MMS-1B has the lowest WER and CER against both annotators (0.76/0.39 and 0.70/0.36), followed by fine-tuned wav2vec2; the whole Whisper family, including the proposed joint-data system, falls behind both. Against the first annotator the three Whisper training configurations are effectively tied (0.92–0.93), so the CS/BN/ALL ordering seen on the silver benchmark does not survive a human reference. This ranking change is important evidence of reference-dependent bias and is reported as a central result rather than a secondary caveat.

The fine-tuned Whisper systems have lower error than zero-shot Whisper-small under both human references, and MMS-1B's result motivates a larger adjudicated evaluation across model families. Because the current sample contains only 100 clips and two independent, non-adjudicated transcripts, the human ranking is informative but not definitive; this is a limited validation study rather than a conclusive model-ranking experiment.

I. LLM-Ensemble Diagnostic

After the human-reference evaluation, we include a secondary text-only diagnostic using Mistral-7B [35], LLaMA-3.1-8B [36], and Gemma-3-12B [37]. The judges score the same fixed, released 50-clip list (25 transcript-labeled CS and 25 Bengali-only clips) for semantic plausibility, apparent code-switch preservation, and dialect naturalness. They see neither audio nor a reference transcript.

BN_ONLY has the highest nominal overall score (2.89/5), followed by ALL (2.81), CS_ONLY (2.68), and large-v3 (2.47); the ALL–BN_ONLY paired difference is not detectable on this sample ($p = 0.1532$). More importantly, BN_ONLY receives the highest apparent code-switch score despite an EN-Surv of only 0.01. This mismatch demonstrates that the judges assess textual plausibility, not fidelity to spoken English tokens. The diagnostic is not a human mean-opinion score and does not override the human-reference, WER, EN-Surv, or CS-F1 results.

TABLE XVII
Secondary text-only LLM diagnostic (mean score 0–5) on one shared 50-clip sample. "Agree" is the mean fraction of judge pairs within one point. These scores measure output plausibility, not acoustic fidelity.

Model	Sem.	CS-pres.	Dialect	Overall	Agree
Whisper large-v3 (zs)	2.40	2.43	2.57	2.47	0.713
Whisper BN_ONLY (A)	2.91	3.03	2.73	2.89	0.811
Whisper CS_ONLY (B)	2.74	2.73	2.56	2.68	0.789
Whisper ALL (joint)	2.82	2.92	2.68	2.81	0.787

VIII. Limitations

Several limitations bound the claims of this study. First, the transcripts are silver-standard: they are generated by Whisper large-v3 and textually filtered by Qwen2.5, which cannot inspect the audio. The two-annotator validation on 100 clips estimates silver-label WER/CER at 0.73/0.44 and 0.80/0.49 against the two independent references, but it is neither large nor adjudicated. Because Whisper generated the silver references, rankings may also favor Whisper-like outputs. The Qwen cascade is reproducible and its retention rate is reported, but its accept/reject accuracy has not been independently human-labeled. The archived construction snapshot stores overall Qwen votes but not all four criterion values; the released script now records those values and derives the decision deterministically. A regenerated release should therefore be versioned rather than assumed to be byte-identical to the archived snapshot.

Second, the script-based detector labels 2,361 of 41,499 silver transcripts (5.7%) as code-switched, with only 176 positives in the test set. The reported bootstrap intervals expose the resulting uncertainty, and the CS findings should not be generalized beyond Bengali–English switching in these social-media domains. Third, the baseline suite is representative rather than exhaustive. Larger fine-tuned models and Bengali-capable SeamlessM4T may change absolute rankings; Canary-1B is not a Bengali baseline. No cross-corpus evaluation on Ben-10, Common Voice, or another external Bengali test set was run, so this study makes no claim of out-of-domain generalization.

The archived historical split is also not source-disjoint. Sixty-nine of 101 inferred test sources occur elsewhere in the retained corpus, and 326 exact- audio duplicate groups cross the test boundary. Consequently, fine-tuned model scores cannot be interpreted as leakage-free generalization. A repaired split is frozen, but the systems require retraining before it can support new model comparisons. Within the historical test set, clips from the same source video are dependent; clip-level bootstrap intervals may therefore be too narrow. Released source-cluster analyses address that unit of dependence only partially, because source IDs neither prove speaker independence nor repair cross-split contamination.

The large-v3 reference/evaluation discrepancy is likewise not fully reconstructible. With construction-time transcripts as references, the paired evaluation-time outputs differ at 0.927 WER; against the stored silver references, those same outputs obtain 0.930 WER. The current code shows different inference backends and decoding settings, but missing historical revisions, software versions, and paired audio hashes prevent a unique causal diagnosis. Current-environment details and available artifact hashes are archived separately in REFERENCE_PIPELINE_PROVENANCE.md; they are not attributed to the historical runs.

Finally, dialect and source domain are the only available audit dimensions. The absence of consented demographic metadata prevents age-, gender-, socioeconomic-, and urban–rural fairness analyses. Search and platform selection may overrepresent more visible, digitally connected speakers, and the dialect distribution is imbalanced. The informal vlog, comedy, review, and diaspora domains also differ from broadcast news, healthcare, and education. The LLM-ensemble semantic scores add a meaning-preservation diagnostic, but they remain model-based and cannot substitute for human judgments; being reference-free and text-only, they also cannot detect content that the model silently omitted, which is why code-switching fidelity is claimed only from the reference-based metrics. These limitations motivate consent-based demographic auditing, targeted CS collection, external-corpus testing, and a larger adjudicated human set when resources permit.

IX. Conclusion

AncholVox contributes a silver-standard diagnostic benchmark rather than a new state-of-the-art ASR model. All model comparisons use a contaminated historical split and therefore do not constitute leakage-free generalization results. On that split, fine-tuned wav2vec2 has the lowest primary silver-reference WER (0.861), whereas the limited 100-clip human-reference study gives MMS-1B the lowest WER against both annotators. The changed ordering demonstrates that the rankings are reference-dependent.

The Whisper recipes describe trade-offs rather than a single winner. Compared with BN_ONLY, ALL has higher transcript-derived CS-F1 (0.192 versus 0.011) and English-token survival (0.13 versus 0.01), but CS_ONLY is higher on both switching measures and has worse overall silver-reference WER. Because these recipes differ in coverage, sample count, and optimization budget, the differences are descriptive associations, not causal estimates of training-data composition. Repetition collapse likewise changes the deployed output condition rather than the acoustic model and is therefore reported separately.

The benchmark is useful for studying silver-reference bias, regional variation, English-token retention, and decoder repetition under transparent validity limits. Priorities for future work are retraining on the repaired source-disjoint split, expanding and adjudicating the human test set, completing rights review, testing external Bengali corpora and Bengali-capable architectures, improving repetition-aware decoding, and conducting consent-based demographic auditing.

Data and Code Availability

The dataset landing page and code are available at https://huggingface.co/datasets/niloycste68/Bangali_local_dialect_ASR_HF_Dataset and https://github.com/niloycste/ASR-Bengali-Local-Dialect-. The public resource name is AncholVox; these legacy URL slugs are retained to avoid breaking existing links. Audio is under gated access: the release register (ETHICS_AND_DATA_GOVERNANCE.md in the code repository) records three blocking determinations — institutional ethics confirmation, redistribution review, and per-source allow/remove decisions — none of which has yet been made, and hosted audio is not publicly downloadable until they are. Describing this procedure is not a claim to have completed it; the register states its status as of the submission date. The repository documents the multi-stage commands, augmentation step, model identifiers, hyperparameters, split settings, fixed statistical seed, dependency lock file, and checkpoint manifest. Released prediction files reproduce the reported metrics, confidence intervals, and two-sided p-values without retraining.

The historical test manifest on which every reported number depends is archived in full: 4,088 clips, 176 transcript-labelled code-switched clips, and 13.2 hours, matching Table IV. Historical train and development manifests are absent, and the audit shows that the historical test set is not source-disjoint from the retained remainder. We therefore do not describe the archived scores as leakage-free. For future retraining, the construction code now sorts inferred sources before seeded assignment, and a separate repaired split is frozen as original-clip manifests with SHA-256 checksums under dataset_pipeline/repaired_split_manifests. The historical and repaired partitions must not be conflated; no result from the repaired split is reported in this paper. End-to-end checkpoint reuse additionally requires depositing the large trained weights at the archived locations listed in the manifest. Audio redistribution is governed by the rights-review procedure described in Section III; where redistribution is not authorized, the release is limited to permissible metadata, source pointers, and processing code.

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